

1. GROUP BY kya hota hai?

Definition

GROUP BY SQL clause ka use rows ko kisi ek ya multiple columns ki same values ke basis par groups mein divide karne ke liye hota hai.

GROUP BY ka actual fayda tab hota hai jab hum **aggregate functions** use karte hain:

```
COUNT()  
SUM()  
AVG()  
MIN()  
MAX()
```



 **SQL**

```
SELECT city, COUNT(*) AS total_students  
FROM student  
GROUP BY city;
```

Yahan SQL:

Student table



Same city wale students ko group karo



Har group ka COUNT nikalo

</> SQL

```
SELECT city, COUNT(*) AS total_students  
FROM student  
GROUP BY city;
```

to SQL internally roughly aise groups banayega:

Indore

- |— Aarav
- |— Krishna
- |— Ananya
- |— Rahul
- |— ...
- └— Chirag

Bhopal

- |— Vivaan
- |— Sarthak
- |— Kavya
- |— ...
- └— Kriti



5. GROUP BY + AVG()

Question:

Har city ke students ke average marks kya hain?

SQL

```
SELECT
    city,
    AVG(marks) AS average_marks
FROM student
GROUP BY city;
```

Yahan:

```
Indore → Indore students ka average
Bhopal → Bhopal students ka average
Ujjain → Ujjain students ka average
```

6. GROUP BY + MAX()

Question:

Har department mein highest marks kya hain?

</> SQL

```
SELECT
    dept_id,
    MAX(marks) AS highest_marks
FROM student
GROUP BY dept_id;
```

8. GROUP BY + SUM()

</> SQL

```
SELECT
    dept_id,
    SUM(marks) AS total_marks
FROM student
GROUP BY dept_id;
```

Har department ke students ke marks ka total milega.

9. Multiple aggregate functions

Interview mein ye bahut common hai:

`</> SQL`

```
SELECT
    dept_id,
    COUNT(*) AS total_students,
    AVG(marks) AS avg_marks,
    MAX(marks) AS highest_marks,
    MIN(marks) AS lowest_marks,
    SUM(marks) AS total_marks
FROM student
GROUP BY dept_id;
```

Ek query mein department-wise:

- students count
- average marks
- highest marks
- lowest marks

10. HAVING kya hota hai?

Ab main important part.

Definition

HAVING ka use GROUP BY se banaye gaye groups ko filter karne ke liye hota hai.

Simple language:

WHERE → rows filter
HAVING → groups filter



Sirf woh cities dikhao jahan 10 se zyada students hain.

</> SQL

```
SELECT
    city,
    COUNT(*) AS total_students
FROM student
GROUP BY city
HAVING COUNT(*) > 10;
```

Flow:

```
student
  ↓
GROUP BY city
  ↓
city-wise groups
  ↓
COUNT(*)
  ↓
HAVING COUNT(*) > 10
  ↓
sirf qualifving cities
```



12. WHERE vs HAVING — सबसे important 🔥

WHERE

Individual rows ko filter karta hai.

Example:

</> SQL

```
SELECT *  
FROM student  
WHERE marks > 80;
```

Meaning:

Mujhe 80 se zyada marks wale **students** chahiye.

Question:

✦ Upgrade

Har city mein 80 ya usse zyada marks wale students count karo, aur sirf woh cities dikhao jahan aise students 3 se zyada hain.

</> SQL



```
SELECT  
    city,  
    COUNT(*) AS total_students  
FROM student  
WHERE marks >= 80  
GROUP BY city  
HAVING COUNT(*) > 3;
```

Execution samjho:

```
1. FROM student
   ↓
2. WHERE marks >= 80
   ↓
3. GROUP BY city
   ↓
4. COUNT(*)
   ↓
5. HAVING COUNT(*) > 3
   ↓
6. Result
```

Difference:

Difference:

`</>` SQL

`WHERE marks >= 80`

→ students filter

`</>` SQL

`HAVING COUNT(*) > 3`

→ city groups filter

Question.

Department-wise total students dikhao.

`</>` SQL

```
SELECT
    d.dept_name,
    COUNT(s.id) AS total_students
FROM department d
JOIN student s
    ON d.dept_id = s.dept_id
GROUP BY d.dept_name;
```

Yahan:

```
department
  +
student
  ↓
JOIN
  ↓
Department name + student
  ↓
```



16. Department-wise average marks

</> SQL

```
SELECT
    d.dept_name,
    AVG(s.marks) AS avg_marks
FROM department d
JOIN student s
    ON d.dept_id = s.dept_id
GROUP BY d.dept_name;
```

Sirf woh departments dikhao jinka average marks 80 se zyada hai.

</> SQL

```
SELECT
    d.dept_name,
    AVG(s.marks) AS avg_marks
FROM department d
JOIN student s
    ON d.dept_id = s.dept_id
GROUP BY d.dept_name
HAVING AVG(s.marks) > 80;
```

Yahan:

</> SQL

```
GROUP BY d.dept_name
```

department-wise groups bana raha hai.

19. Gender-wise student count

Question:

Male aur Female students kitne hain?

SQL

```
SELECT
    gender,
    COUNT(*) AS total_students
FROM student
GROUP BY gender;
```

20. Course year-wise student count

SQL

```
SELECT
    course_year,
    COUNT(*) AS total_students
FROM student
GROUP BY course_year;
```

Result conceptually:

```
1st year → count
2nd year → count
3rd year → count
4th year → count
```



22. Female students ka city-wise count

Yahan `WHERE + GROUP BY` :

`</>` SQL

```
SELECT
    city,
    COUNT(*) AS female_students
FROM student
WHERE gender = 'Female'
GROUP BY city;
```

Flow:

```
All students
  ↓
WHERE gender = Female
  ↓
Only female students
  ↓
GROUP BY city
  ↓
City-wise female count
```



23. Female students wale cities jahan 5 se zyada females hain

🔥 Interview-level:

```
</> SQL   
  
SELECT  
    city,  
    COUNT(*) AS female_students  
FROM student  
WHERE gender = 'Female'  
GROUP BY city  
HAVING COUNT(*) > 5;
```

WHERE

↓

Female rows select

GROUP BY

↓

City-wise group

HAVING

↓

5 se zyada female wale groups

24. Highest average marks wala department

Ye thoda advanced/interview level hai:

`</>` SQL

```
SELECT
    d.dept_name,
    AVG(s.marks) AS avg_marks
FROM department d
JOIN student s
    ON d.dept_id = s.dept_id
GROUP BY d.dept_name
ORDER BY avg_marks DESC
LIMIT 1;
```

GROUP BY department-wise average nikalta hai.

ORDER BY highest average ko upar lata hai.

LIMIT 1 top department deta hai.



25. Second highest average department

</> SQL

```
SELECT
    d.dept_name,
    AVG(s.marks) AS avg_marks
FROM department d
JOIN student s
    ON d.dept_id = s.dept_id
GROUP BY d.dept_name
ORDER BY avg_marks DESC
LIMIT 1 OFFSET 1;
```

LIMIT 1 → 1 row
OFFSET 1 → first row skip

✦ Upgrade



26. Department-wise highest marks

</> SQL



```
SELECT
    d.dept_name,
    MAX(s.marks) AS highest_marks
FROM department d
JOIN student s
    ON d.dept_id = s.dept_id
GROUP BY d.dept_name;
```

Important: Isse highest marks milega, lekin highest marks wale **student ka naam** directly nahi milega.

Har department mein gender-wise students count karo.

`</> SQL`

```
SELECT
    dept_id,
    gender,
    COUNT(*) AS total_students
FROM student
GROUP BY dept_id, gender;
```

Grouping:

```
101 + Male
101 + Female

102 + Male
102 + Female

103 + Male
103 + Female
...
```



15% Salary Decrease

</> SQL

```
UPDATE employee  
SET salary = salary * 0.85;
```

Example:

Old Salary = 50,000

15% decrease = $50,000 \times 15 / 100 = 7,500$

New Salary = 42,500



WHERE ke saath

Sirf CSE employees ki salary 10% increase:

</> SQL

```
UPDATE employee
SET salary = salary * 1.10
WHERE dept_id = 101;
```

Sirf jiski salary ₹50,000 se kam hai uski 15% decrease:

</> SQL

```
UPDATE employee
SET salary = salary * 0.85
WHERE salary < 50000;
```

 Shortcut

Operation	Formula
10% increase	<code>salary * 1.10</code>
20% increase	<code>salary * 1.20</code>
15% increase	<code>salary * 1.15</code>
10% decrease	<code>salary * 0.90</code>
15% decrease	<code>salary * 0.85</code>
20% decrease	<code>salary * 0.80</code>

Rule:

Increase `X%` → `salary * (1 + X/100)`

Decrease `X%` → `salary * (1 - X/100)`